

# “Luckily, we’re not getting a robot like this”: Co-designing household robot privacy with t(w)eens and their families

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## Research Questions

- What are t(w)eens’ privacy concerns about and preferred privacy designs for household robots?
- How do t(w)eens’ perceptions and preferences compare to their parents?

## Background

As household robots become increasingly common<sup>1,2</sup>, researchers have started exploring users’:

- Privacy concerns around robots<sup>3</sup>
- Privacy design preferences for robots<sup>4</sup>

Most studies focus on adults from single-user perspectives

## Sample & Design

**Sample:** 8 U.S.-based children between 8-17 y/o ( $M=13$ ,  $SD=3.02$ ) and their parent(s) (6 mothers & 2 fathers, 100% 4-yr degree or higher, 66.67% annual income >80k)

6 families total; 2 families w/ two parents (205, 211), 2 families with sibling(s) (206, 218), 2 families w/ one parent (230, 234)

**Design:** 120 min co-design sessions in living room lab conducted between July-August 2025

- At least one parent/child present during interview
- Meet and greet with Temi V3 robot
- Activities:
  - Designate areas of household where robot can/cannot have access on home blueprint
  - Generate privacy designs based on scenarios involving robot
  - Review designs, household access blueprint

**Analysis:** Inductive thematic analysis<sup>5</sup> by first 3 authors

(in results, P= parent, C= Child, #= family ID; P#\_1 or P#\_2 if two parents, C#\_1, C#\_2, or C#\_3 if siblings)

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## References

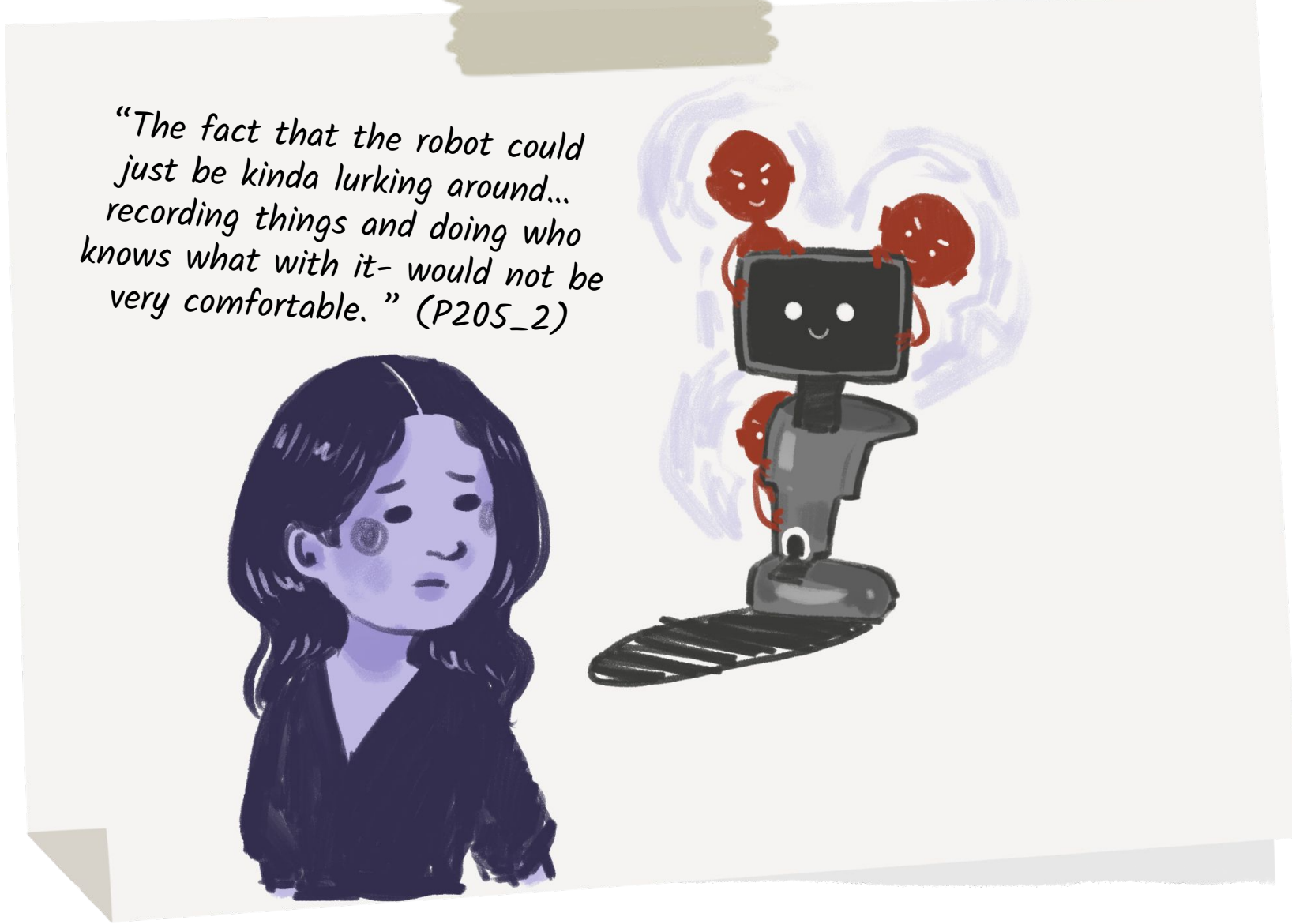
<sup>1</sup> IX Technologies. (2025). NEO Home Robot. <https://www.ix.tech/neo>  
<sup>2</sup> Figure AI. (2025). Introducing Figure 03. <https://www.figure.ai/news/introducing-figure-03>  
<sup>3</sup> Levinson, L., Barrett, T., Gomez, R., & Sabanovic, S. (2024). Surveying adult perceptions of privacy and attitudes towards social robots in the home. In Companion of the 2024 ACM/IEEE International Conference on Human-Robot Interaction, 664-668. <https://doi.org/10.1145/3610978.3649719>  
<sup>4</sup> Bell, H., Kwesi, J., Laabadi, H., & Emami-Naeini, P. (2025). "Is it always watching? Is it always listening?" Exploring contextual privacy and security concerns around domestic social robots. *arXiv preprint arXiv:2508.10786*.  
<sup>5</sup> Clarke, V., & Braun, V. (2014). Thematic analysis. In Encyclopedia of critical psychology (pp. 1947-1952). Springer, New York, NY.

## Takeaway

T(w)eens were concerned about how a household robot might fit into in their homes, and wanted privacy designs that would offer them some autonomy over a) how the robot operated around them and b) what data the robot could share with their parents.

Overall, t(w)een and adult participants raised similar privacy issues, but emphasized different concerns.

While parents more often discussed feeling concerned about unauthorized data access or use by hackers or the robot’s manufacturers...



... t(w)eens were more likely to focus on their discomfort with the robot’s physical features and the robot’s lack of functional purpose.

Some t(w)eens and parents disagreed over...

areas of their home the robot could access

how the robot should be designed



T(w)eens’ wanted overt privacy designs that allowed some control over if/how their personal data was being recorded and shared.

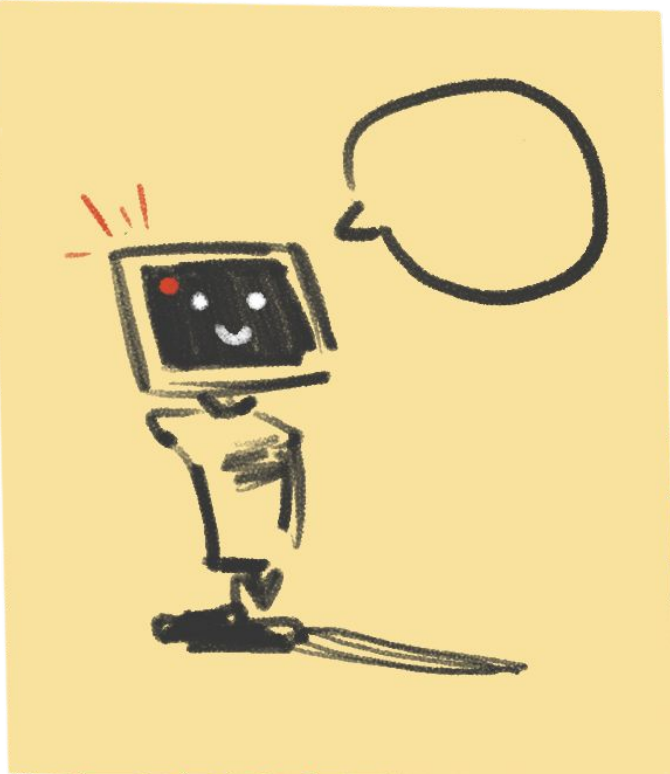
However, multiple t(w)eens expected that their parents would have ultimate control over the robot, and might use the robot to monitor them.

Camera Controls

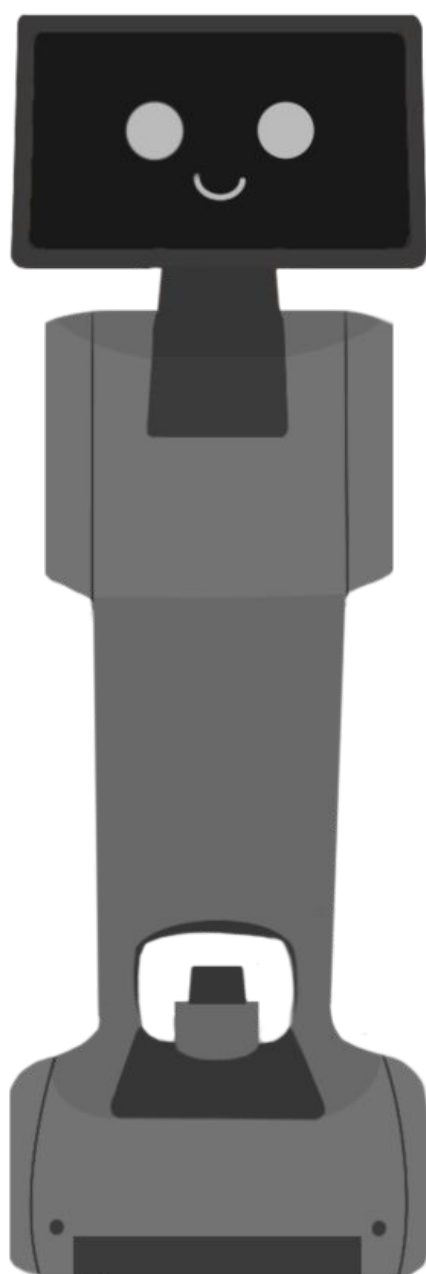


"It could have this thing where it can't look at screens unless you may tell it to look at it." (C206\_2)

Verbal/ Visual Notifications



"[I would want]... both, like an audible and a visual... Probably a transcript on what you say... or, like a noise. Like 'ding, ding.'" (C218)

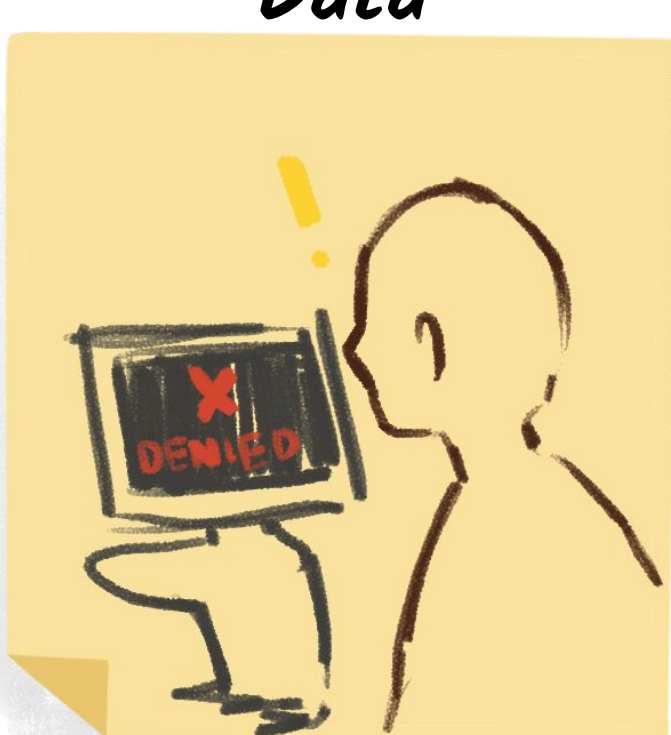


Autonomous Learning



"Because I don't want to have to go through and say... 'when we say this, it's sensitive'... We want to have a certain level of control, but a more higher level [of control]." (C211)

User Control of Data



"I would say there would be a password that only you and the adults know [and not your siblings] if it's your account." (C206\_3)